

Variations on Logistic Regression

4.1 Latent Variables

Suppose that students answer questions on a test and that a specific student has an aptitude T . A particular question might have difficulty d and the student will get the answer correct only if $T > d$. Now if we consider d fixed and T as a random variable with density f and distribution function F , then the probability that the student will get the answer wrong is:

$$p = P(T \leq d) = F(d)$$

T is called a *latent variable*. Suppose that the distribution of T is *logistic*:

$$F(y) = \frac{\exp(y - \mu)/\sigma}{1 + \exp(y - \mu)/\sigma}$$

So

$$\text{logit}(p) = -\mu/\sigma + d/\sigma$$

If we set $\beta_0 = -\mu/\sigma$ and $\beta_1 = 1/\sigma$, we now have a logistic regression model. We can illustrate this in the following example where we set $d = 1$ and let T have mean -1 and $\sigma = 1$:

```
x <- seq(-6, 4, 0.1)
y <- dlogis(x, location=-1)
plot(x, y, type="l", ylab="density", xlab="t")
ii <- (x <= 1)
polygon(c(x[ii], 1, -6), c(y[ii], 0, 0), col='gray')
```

The plot in Figure 4.1 shows a logistically distributed latent variable. We can see that this distribution is apparently very similar to the normal distribution. The shaded area represents the probability of getting an answer wrong. As the mean aptitude of this student is somewhat less than the difficulty of the question, this probability is substantially greater than one half.

This idea also arises in a *bioassay* where we might treat an animal, plant or person with some concentration of a treatment and observe the outcome. For example, suppose we are interested in the concentration of insecticide to be used in exterminating a pest. Insects will have varying tolerances for the toxin and will survive if their tolerance is greater than the dose. In this context, the term *tolerance distribution* for T is used. Applications in several other areas exist where we observe only a binary outcome but believe this to be generated by some continuous but unobserved variable.

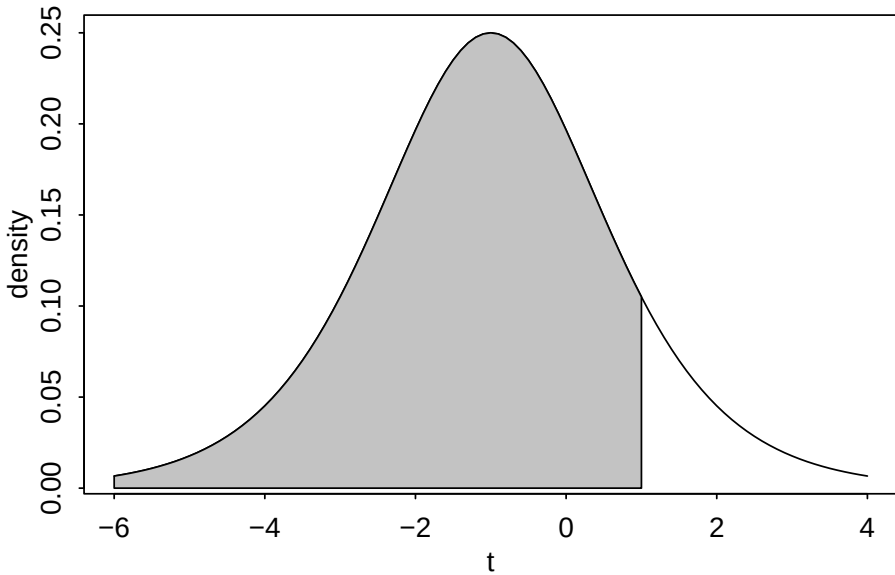


Figure 4.1 *Probability of getting the answer wrong for logistic latent variable.*

4.2 Link Functions

Until now we have used logit link function to connect the probability and the linear predictor. But other choices of link function are reasonable. We need a function that bounds the probability between zero and one. We also expect the link function to be monotone. It is conceivable that the success probability may go up and down as the linear predictor increases but this circumstance is best modeled by adding nonlinear components to the linear predictors such as quadratic terms rather than modifying the link function. The latent variable formulation suggests some other possibilities for the link function. Here are some choices which are implemented in the `glm()` function:

1. Probit: $\eta = \Phi^{-1}(p)$ where Φ is the normal cumulative distribution function. This arises from a normally distributed latent variable.
2. Complementary log-log: $\eta = \log(-\log(1-p))$. A Gumbel-distributed latent variable will lead to this.
3. Cauchit: $\eta = \tan^{-1}(\pi(p - 1/2))$ which is motivated by a Cauchy-distributed latent variable.

We can illustrate the choices using some data from Bliss (1935) on the numbers of insects dying at different levels of insecticide concentration. We fit all four link functions:

```
data(bliss, package="faraway")
```

```
bliss
```

```
  dead alive conc
1    2    28    0
```

```
2      8      22      1
3     15     15      2
4     23      7      3
5     27      3      4
```

```
mlogit <- glm(cbind(dead,alive) ~ conc, family=binomial, data=bliss)
mprobit <- glm(cbind(dead,alive) ~ conc, family=binomial(link=probit),
  ↪ data=bliss)
mcloglog <- glm(cbind(dead,alive) ~ conc, family=binomial(link=cloglog
  ↪ ), data=bliss)
mcauchit <- glm(cbind(dead,alive) ~ conc, family=binomial(link=cauchit
  ↪ ), data=bliss)
```

We start by considering the fitted values:

```
fitted(mlogit)
      1      2      3      4      5
0.089172 0.238323 0.500000 0.761677 0.910828
```

or from `predict(mlogit, type="response")`. These are constructed using linear predictor, η :

```
coef(mlogit)[1]+coef(mlogit)[2]*bliss$conc
[1] -2.3238 -1.1619 0.0000 1.1619 2.3238
```

Alternatively, these values may be obtained from `modl$linear.predictors` or `predict(modl)`. The fitted values are then:

```
library(faraway)
ilogit(mlogit$lin)
      1      2      3      4      5
0.089172 0.238323 0.500000 0.761677 0.910828
```

Notice the need to distinguish between predictions in the scale of the response and the link. Now compare the logit, probit and complementary log-log fits:

```
predval <- sapply(list(mlogit,mprobit,mcloglog,mcauchit),fitted)
dimnames(predval) <- list(0:4,c("logit","probit","cloglog","cauchit"))
round(predval,3)
      logit probit cloglog cauchit
0 0.089 0.084 0.127 0.119
1 0.238 0.245 0.250 0.213
2 0.500 0.498 0.455 0.506
3 0.762 0.752 0.722 0.791
4 0.911 0.914 0.933 0.882
```

These are not very different, but now look at a wider range from $[-4, 8]$. We apply the `predict` function to each of the four models forming a matrix of predicted values. We label the columns and add the information about the dose. The `tidyr` package is useful for reformatting data from a wide format of multiple measured values per row to a long format where there is only one response value per row. This is accomplished using the `gather()` function. This format, where each row is an observation and each column is a variable, is the most convenient form for many R analyses. Finally, the `ggplot2` package is useful for completing and well-labeled plot.

```
dose <- seq(-4,8,0.2)
predval <- sapply(list(mlogit,mprobit,mcloglog,mcauchit), function(m)
  ↪ predict(m,data.frame(conc=dose), type="response"))
colnames(predval) <- c("logit","probit","cloglog","cauchit")
predval <- data.frame(dose,predval)
library(tidyr)
mpv <- gather(predval, link, probability, -dose)
library(ggplot2)
ggplot(mpv, aes(x=dose,y=probability,linetype=link))+geom_line()
```

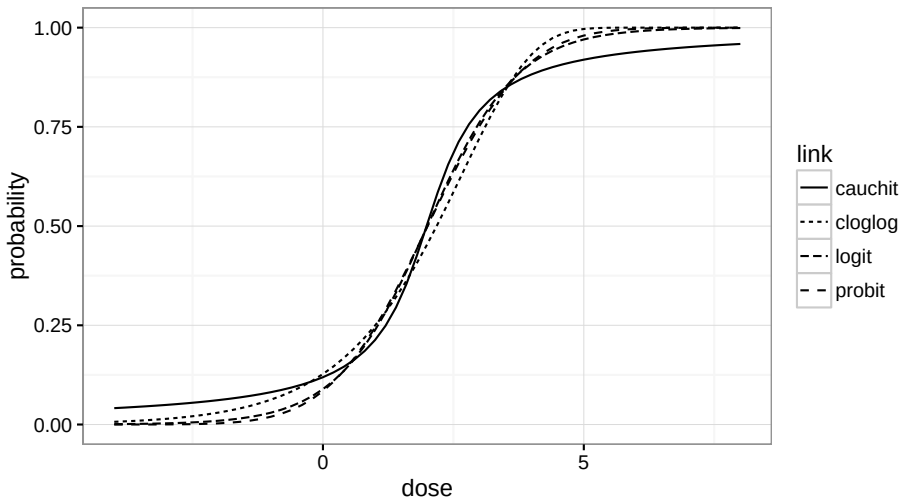


Figure 4.2 *Probit, logit, complementary log-log and cauchit compared. The data range from 0 to 4. We see that the links are similar in this range and only begin to diverge as we extrapolate.*

The predicted probabilities as shown in Figure 4.2 do not differ much in the range of the data but the differences become more apparent as we extrapolate. The cauchit link is most different as the corresponding latent variable is most variable and would be slowest to converge to zero on the left and one on the right. The complementary log-log can also be clearly distinguished from the logit and probit. But the logit and probit are harder to separate.

The difference between the probit and the logit becomes apparent when we examine the ratio of the probabilities in the lower and upper tails:

```
ggplot(predval, aes(x=dose, y=probit/logit))+geom_line()+xlim(c(-4, 0))
ggplot(predval, aes(x=dose, y=(1-probit)/(1-logit)))+geom_line()+xlim(c(
  4, 8))
```

In Figure 4.3, we see that the probit and logit differ substantially in the tails. The same phenomenon is observed for the complementary log-log. This is problematic since the former plot indicates it would be difficult to distinguish between the two using the data we have. This is an issue in trials of potential carcinogens and other substances that must be tested for possible harmful effects on humans. Some substances are highly poisonous in that their effects become immediately obvious at doses that might normally be experienced in the environment. It is not difficult to detect such substances. However, there are other substances whose harmful effects only become apparent at large dosages where the observed probabilities are sufficiently larger than zero to become estimable without immense sample sizes. In order to estimate the probability of a harmful effect at a low dose, it would be necessary to select an appropriate link function and yet the data for high dosages will be of little help in doing this. As Paracelsus (1493–1541) said, “All substances are poisons; there is none which is not a poison. The right dose differentiates a poison.”

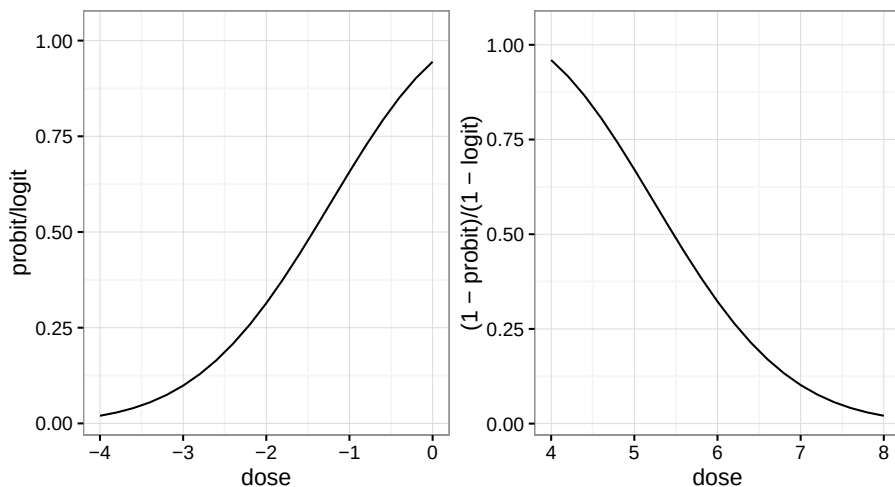


Figure 4.3 *Ratio of predicted probit to logit probabilities in the lower tail on the left and in the upper tail to the right.*

A good example of this problem is asbestos. Information regarding the harmful effects of asbestos derives from historical studies of workers in industries exposed to very high levels of asbestos dust. However, we would like to know the risk to individuals exposed to low levels of asbestos dust such as those found in old buildings. It is very difficult to accurately determine this risk. We cannot accurately measure exposure or outcome. This is not to argue that nothing should be done, but that decisions should be made in recognition of the uncertainties.

We must choose a link function to specify a binomial regression model. It is usually not possible to make this choice based on the data alone. For regions of moderate p , that is not close to zero or one, the link functions we have proposed are quite similar and so a very large amount of data would be necessary to distinguish between them. Larger differences are apparent in the tails, but for very small p , one needs a very large amount of data to obtain just a few successes, making it expensive to distinguish between link functions in this region. So usually, the choice of link function is made based on assumptions derived from physical knowledge or simple convenience.

The default choice is the logit link. There are three advantages: it leads to simpler mathematics than the probit due to the intractability of Φ , it is easier to interpret using odds and it allows easier analysis of retrospectively sampled data as we shall see in the next section.

4.3 Prospective and Retrospective Sampling

Consider the data shown in Table 4.1 from a study on infant respiratory disease which shows the proportions of children developing bronchitis or pneumonia in their first year of life by type of feeding and sex, which may be found in Payne (1987):

	Bottle Only	Some Breast with Supplement	Breast Only
Boys	77/458	19/147	47/494
Girls	48/384	16/127	31/464

Table 4.1 *Incidence of respiratory disease in infants to the age of 1 year.*

We can recover the layout above with the proportions as follows:

```
data(babyfood, package="faraway")
xtabs(disease/(disease + nondisease) ~ sex + food, babyfood)
```

```
      food
sex   Bottle  Breast  Suppl
Boy  0.16812 0.095142 0.12925
Girl 0.12500 0.066810 0.12598
```

In *prospective* sampling, the predictors are fixed and then the outcome is observed. This is also called a *cohort study*. In the infant respiratory disease example shown in Table 4.1, we would select a sample of newborn girls and boys whose parents had chosen a particular method of feeding and then monitor them for their first year.

In *retrospective* sampling, the outcome is fixed and then the predictors are observed. This is also called a *case-control study*. Typically, we would find infants coming to a doctor with a respiratory disease in the first year and then record their sex and method of feeding. We would also obtain a sample of respiratory disease-free infants and record their information. The method for obtaining the samples is important — we require that the probability of inclusion in the study is independent of the predictor values.

Since the question of interest is how the predictors affect the response, prospective sampling seems to be required. Let's focus on just boys who are breast or bottle fed. The data we need is:

```
babyfood[c(1, 3), ]
  disease nondisease sex  food
1      77         381 Boy Bottle
3      47         447 Boy Breast
```

As we have seen in Section 2.2, the log-odds is a sensible way to measure the strength of the association of a predictor with the outcome.

- Given the infant is *breast* fed, the log-odds of having a respiratory disease are $\log 47/447 = -2.25$.
- Given the infant is *bottle* fed, the log-odds of having a respiratory disease are $\log 77/381 = -1.60$.

The difference between these two log-odds, $\Delta = -1.60 - -2.25 = 0.65$, represents the increased risk of respiratory disease incurred by bottle feeding relative to breast feeding. This is the log-odds ratio. In this case, the log-odds ratio is positive indicating a greater risk of respiratory disease for bottle-fed compared to breast-fed babies.

Now suppose that this had been a retrospective study — we could compute the log-odds of feeding type given respiratory disease status and then find the difference.

Notice that this would give the same result because:

$$\Delta = \log 77/47 - \log 381/447 = \log 77/381 - \log 47/447 = 0.65$$

This shows that a retrospective design is as effective as a prospective design for estimating Δ . This manipulation is not possible for other links such as the probit so we have to use the logit.

Retrospective designs have several advantages compared to prospective designs. In this example, we would need to wait a year before the outcome for a particular baby is known when using the cohort approach. For many other responses such as cancer or heart disease and even nonmedical outcomes related to career achievements, we may need to wait a long time before the results are known. In contrast, the case-control study can be done quickly. Also, we can investigate many possible predictors using the retrospective approach whereas in the prospective study, we must specify these in advance and we may not be gifted with foresight. Retrospective studies are particularly valuable for rare outcomes, otherwise a very large cohort will be necessary to guarantee any “positive” responses.

Prospective designs have other advantages. They are less susceptible to bias in the selection of the sample. Retrospective studies rely on historical records which may be of unknown accuracy and completeness. They may also rely on the memory of the subject which may be unreliable. Prospective studies also allow the study of more than one outcome. For example, we might also measure babies’ success in learning to walk. This is problematic in the case-control approach because more than one outcome would lead to different samples. We shall also see that only prospective studies can generate models that predict the probability of an outcome.

In most practical situations, we will also need to account for the effects of covariates X . Let π_0 be the probability that an individual is included in the study if they do *not* have the disease, while let π_1 be the probability of inclusion if they do have the disease. For a prospective study, $\pi_0 = \pi_1$ because we have no knowledge of the outcome, while for a retrospective study typically π_1 is much greater than π_0 . Suppose that for given x , $p^*(x)$ is the conditional probability that an individual has the disease given that he or she was included in the study, while $p(x)$ is the unconditional probability that he or she has the disease as we would obtain from a prospective study. Now by Bayes theorem:

$$p^*(x) = \frac{\pi_1 p(x)}{\pi_1 p(x) + \pi_0 (1 - p(x))}$$

which can be rearranged to show that:

$$\text{logit}(p^*(x)) = \log \frac{\pi_1}{\pi_0} + \text{logit}(p(x))$$

So the only difference between the retrospective and the prospective study would be the difference in the intercept: $\log(\pi_1/\pi_0)$. Generally π_1/π_0 would not be known, so we would not be able to estimate β_0 , but knowledge of the other β would be most important since this can be used to assess the *relative* effect of the covariates.

We could not, however, estimate the absolute effect. Hence, a retrospective study might tell us that a particular lifestyle choice would double the risk of an unpleasant outcome but it cannot tell us the probability of that outcome. If the probability is not high, we may not care.

4.4 Prediction and Effective Doses

Sometimes we wish to predict the outcome for given values of the covariates. For binomial data this will mean estimating the probability of success. Given covariates x_0 , the predicted response on the link scale is $\hat{\eta} = x_0\hat{\beta}$ with variance given by $x_0^T(X^TWX)^{-1}x_0$. Approximate confidence intervals may be obtained using a normal approximation. To get an answer in the probability scale, it will be necessary to transform back using the inverse of the link function. We predict the response for the insect data:

```
data(bliss, package="faraway")
lmod <- glm(cbind(dead,alive) ~ conc, family=binomial,data=bliss)
lmodsum <- summary(lmod)
```

We show how to predict the response at a dose of 2.5:

```
x0 <- c(1,2.5)
eta0 <- sum(x0*coef(lmod))
ilogit(eta0)
```

```
[1] 0.64129
```

A 64% predicted chance of death at this dose — now compute a 95% confidence interval (CI) for this probability. First, extract the variance matrix of the coefficients:

```
(cm <- lmodsum$cov.unscaled)
              (Intercept)      conc
(Intercept)  0.174630 -0.065823
conc         -0.065823  0.032912
```

The standard error on the logit scale is then:

```
se <- sqrt( t(x0) %*% cm %*% x0)
```

so the CI on the probability scale is:

```
ilogit(c(eta0-1.96*se,eta0+1.96*se))
```

```
[1] 0.53430 0.73585
```

A more direct way of obtaining the same result is:

```
predict(lmod,newdata=data.frame(conc=2.5),se=T)
```

```
$fit
```

```
[1] 0.58095
```

```
$se.fit
```

```
[1] 0.2263
```

```
ilogit(c(0.58095-1.96*0.2263,0.58095+1.96*0.2263))
```

```
[1] 0.53430 0.73585
```

Note that in contrast to the linear regression situation, there is no distinction possible between confidence intervals for a future observation and those for the mean response. Now we try predicting the response probability at the low dose of -5 :

```
x0 <- c(1,-5)
se <- sqrt( t(x0) %*% cm %*% x0)
eta0 <- sum(x0*lmod$coef)
ilogit(c(eta0-1.96*se,eta0+1.96*se))
```

```
[1] 2.3577e-05 3.6429e-03
```


This is not a wide interval in absolute terms, but in relative terms, it certainly is. The upper limit is about 100 times larger than the lower limit.

When there is a single (continuous) covariate or when other covariates are held fixed, we sometimes wish to estimate the value of x corresponding to a chosen p . For example, we may wish to determine which dose, x , will lead to a probability of success p . ED50 stands for the *effective dose* for which there will be a 50% chance of success. When the objective is to kill the subjects or determine toxicity, as when using insecticides, the term LD50 would be used. LD stands for *lethal dose*. Other percentiles are also of interest. For a logit link, we can set $p = 1/2$ and then solve for x to find:

$$\widehat{\text{ED50}} = -\hat{\beta}_0/\hat{\beta}_1$$

Using the Bliss data, the LD50 is:

```
(ld50 <- -lmod$coef[1]/lmod$coef[2])
(Intercept)
2
```

To determine the standard error, we can use the delta method. The general expression for the variance of $g(\hat{\theta})$ for multivariate θ is given by

$$\text{var } g(\hat{\theta}) \approx g'(\hat{\theta})^T \text{var } \hat{\theta} g'(\hat{\theta})$$

which, in this example, works out as:

```
dr <- c(-1/lmod$coef[2], lmod$coef[1]/lmod$coef[2]^2)
sqrt(dr %*% lmodsum$cov.un %*% dr) [, ]
[1] 0.17844
```

So the 95% CI is given by:

```
c(2-1.96*0.178, 2+1.96*0.178)
[1] 1.6511 2.3489
```

Other levels may be considered — the effective dose x_p for probability of success p is:

$$x_p = \frac{\text{logit}(p) - \beta_0}{\beta_1}$$

So, for example:

```
(ed90 <- (logit(0.9)-lmod$coef[1])/lmod$coef[2])
(Intercept)
3.8911
```

More conveniently, we may use the `dose.p` function in the `MASS` package:

```
library(MASS)
dose.p(lmod, p=c(0.5, 0.9))
      Dose      SE
p = 0.5: 2.0000 0.17844
p = 0.9: 3.8911 0.34499
```

4.5 Matched Case-Control Studies

In a case-control study, we try to determine the effect of certain risk factors on the outcome. We understand that there are other confounding variables that may affect the outcome. One approach to dealing with these is to measure or record them, include them in the logistic regression model as appropriate and thereby control for

their effect. But this method requires that we model these confounding variables with the correct functional form. This may be difficult. Also, making an appropriate adjustment is problematic when the distribution of the confounding variables is quite different in the cases and controls. So we might consider an alternative where the confounding variables are explicitly adjusted for in the design.

In a *matched case-control study*, we match each case (diseased person, defective object, success, etc.) with one or more controls that have the same or similar values of some set of potential confounding variables. For example, if we have a 56-year-old, Hispanic male case, we try to match him with some number of controls who are also 56-year-old Hispanic males. This group would be called a *matched set*. Obviously, the more confounding variables one specifies, the more difficult it will be to make the matches. Loosening the matching requirements, for example, accepting controls who are 50–60 years old, might be necessary. Matching also gives us the possibility of adjusting for confounders that are difficult to measure. For example, suppose we suspect an environmental effect on the outcome. However, it is difficult to measure exposure, particularly when we may not know which substances are relevant. We could match subjects based on their place of residence or work. This would go some way to adjusting for the environmental effects.

Matched case-control studies also have some disadvantages apart from the difficulties of forming the matched sets. One loses the possibility of discovering the effects of the variables used to determine the matches. For example, if we match on sex, we will not be able to investigate a sex effect. Furthermore, the data will likely be far from a random sample of the population of interest. So although relative effects may be found, it may be difficult to generalize to the population.

Sometimes, cases are rare but controls are readily available. A $1 : M$ design has M controls for each case. M is typically small and can even vary in size from matched set to matched set due to difficulties in finding matching controls and missing values. Each additional control yields a diminished return in terms of increased efficiency in estimating risk factors — it is usually not worth exceeding $M = 5$.

For individual i in the j^{th} matched set, we also observe a covariate vector x_{ij} which will include the risk factors of interest plus any other variables that we may wish to adjust for, but were unable for various reasons to include among the criteria used to match the sets. It is important that the decision to include a subject in the study be independent of the risk factors as in the unmatched case-control studies. Suppose we have n matched sets and that we take $i = 0$ to represent the case and $i = 1, \dots, M$ to represent the controls. We propose a logistic regression model of the following form:

$$\text{logit}(p_j(x_{ij})) = \alpha_j + \beta^T x_{ij}$$

The α_j models the effect of the confounding variables in the j^{th} matched set. Given a matched set j of $M + 1$ subjects known to have one case and M controls, the conditional probability of the observed outcome, or, in other words, that subject $i = 0$ is the case and the rest are controls, is:

$$\frac{\exp \beta^T x_{0j}}{\sum_{i=0}^M \exp \beta^T x_{ij}}$$

Notice that α_j cancels out in this expression. We may then form the conditional likelihood for the model by taking the product over all the matched sets:

$$L(\beta) = \prod_{j=1}^n \left\{ 1 + \sum_{i=1}^M \exp[\beta^T (x_{ij} - x_{0j})] \right\}^{-1}$$

We may now employ standard likelihood methods to make inference — see Breslow (1982) for details. The likelihood takes the same form as that used for the proportional hazards model used in survival analysis. This is convenient because we may use software developed for those models as we demonstrate below. Since the α s are not estimated, we cannot make predictions about individuals, but only make statements about the relative risks as measured by the β s. This same restriction also applies to the unmatched model, so this is nothing new.

In Le (1998), a matched case-control study is presented concerning the association between x-rays and childhood acute myeloid leukemia. The sets are matched on age, race and county of residence. For the most part, there is only one control for each case, but there are a few instances of two controls. We start with a look at the data:

```
data(amlxray, package="faraway")
head(amlxray)
```

	ID	disease	Sex	downs	age	Mray	MupRay	MlowRay	Fray	Cray	CnRay
1	7004	1	F	no	0	no	no	no	no	no	1
2	7004	0	F	no	0	no	no	no	no	no	1
3	7006	1	M	no	6	no	no	no	no	yes	3
4	7006	0	M	no	6	no	no	no	no	yes	2
5	7009	1	F	no	8	no	no	no	no	no	1
6	7009	0	F	no	8	no	no	no	no	no	1

Only the age is presented here as one of the matching variables. In the three sets shown here, we see that both subjects have the same age and the first is the case and the second is the control. The other variables are risk factors of interest.

Downs syndrome is known to be a risk factor. There are only seven such subjects in the dataset:

```
amlxray[amlxray$downs=="yes", 1:4]
```

	ID	disease	Sex	downs
7	7010	1	M	yes
17	7018	1	F	yes
78	7066	1	F	yes
88	7077	1	M	yes
173	7146	1	F	yes
196	7176	1	F	yes
210	7189	1	F	yes

We see that all seven subjects are cases. If we include this variable in the regression, its coefficient is infinite. Given this and the prior knowledge, it is simplest to exclude all these subjects and their associated matched subjects:

```
(ii <- which(amlxray$downs=="yes"))
```

```
[1] 7 17 78 88 173 196 210
```

```
rاملxray <- املxray[-c(ii,ii+1),]
```

The variables Mray, MupRay and MlowRay record whether the mother has ever had an x-ray, ever had an upper body x-ray and ever had a lower body x-ray, respectively.

These variables are closely associated, so we will pick just `Mray` for now and investigate the others more closely if indicated. We will also use `CnRay`, a four-level ordered factor grouping the number of x-rays that the child has received in preference to `Cray` which merely indicates whether the child has ever had an x-ray.

The `clogit` function fits a conditional logit model. Since the likelihood is identical with that from a proportional hazards model, it may be found in the `survival` package. The matched sets must be designated by the `strata` function:

```
library(survival)
cmmod <- clogit(disease ~ Sex+Mray+Fray+CnRay+strata(ID), ramlxray)
summary(cmmod)
```

	coef	exp(coef)	se(coef)	z	p
SexM	0.156	1.17	0.386	0.405	0.6900
Mrayyes	0.228	1.26	0.582	0.391	0.7000
Frayyes	0.693	2.00	0.351	1.974	0.0480
CnRay.L	1.941	6.96	0.621	3.127	0.0018
CnRay.Q	-0.248	0.78	0.582	-0.426	0.6700
CnRay.C	-0.580	0.56	0.591	-0.982	0.3300

	exp(coef)	exp(-coef)	lower .95	upper .95
SexM	1.17	0.855	0.549	2.49
Mrayyes	1.26	0.796	0.401	3.93
Frayyes	2.00	0.500	1.005	3.98
CnRay.L	6.96	0.144	2.063	23.51
CnRay.Q	0.78	1.281	0.249	2.44
CnRay.C	0.56	1.786	0.176	1.78

```
Rsquare= 0.089 (max possible= 0.499 )
Likelihood ratio test= 20.9 on 6 df, p=0.00192
Wald test = 14.5 on 6 df, p=0.0246
Score (logrank) test = 18.6 on 6 df, p=0.0049
```

The overall tests for significance of the predictors indicate that at least some of the variables are significant. We see that `Sex` and whether the mother had an x-ray are not significant. There seems little point in investigating the other x-ray variables associated with the mother. An x-ray on the father is marginally significant. However, the x-ray on the child has the clearest effect. Because this is an ordered factor, we have used linear, quadratic and cubic contrasts. Only the linear effect is significant.

The second table of coefficients gives us information helpful for interpreting the size of the effects. We see that the father having had an x-ray doubles the odds of the disease. The interpretation of the number of x-rays of the child is more difficult to interpret because of the coding. Since we have found only a linear effect, we convert `CnRay` to the numerical values 1–4 using `unclass`. We also drop the insignificant predictors:

```
cmodr <- clogit(disease ~ Fray+unclass(CnRay)+strata(ID), ramlxray)
summary(cmodr)
```

	coef	exp(coef)	se(coef)	z	p
Frayyes	0.670	1.96	0.344	1.95	0.05100
unclass(CnRay)	0.814	2.26	0.237	3.44	0.00058

	exp(coef)	exp(-coef)	lower .95	upper .95
Frayyes	1.96	0.512	0.996	3.84
unclass(CnRay)	2.26	0.443	1.419	3.59

The codes for `Cnray` are 1 = none, 2 = 1 or 2 x-rays, 3 = 3 or 4 x-rays and 4 = 5 or

more x-rays. We see that the odds of the disease increase by a factor of 2.26 as we move between adjacent categories. Notice that the father's x-ray variable is now just insignificant in this regression underlining its borderline status.

An incorrect analysis of this data ignores the matching structure and simply uses a binomial GLM:

```
gmod <- glm(disease ~ Fray+unclass(CnRay), family=binomial, ramlxray)
summary(gmod)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.162	0.301	-3.86	0.00011
Frayyes	0.500	0.308	1.63	0.10405
unclass(CnRay)	0.601	0.177	3.39	0.00071

The results are somewhat different.

Although we have found an effect due to x-rays of the child, we cannot conclude the effect is causal. After all, subjects only have x-rays when something is wrong, so it is quite possible that the x-rays are linked to some unknown causal factor.

Other examples of matched data may be found in Section 6.4.

Exercises

- The Chicago insurance dataset found in `chredlin` concerns the problem of redlining in insurance. Read the `help` page for background. Use `involact` as the response and ignore `volact`.
 - Plot a histogram of the distribution of `involact` taking care to choose the bin width to illustrate the issue with zero values. What fraction of the responses is zero?
 - Fit a Gaussian linear model with `involact` as the response with the other five variables as predictors. Use a log transformation for income. Describe the relationship between these predictors and the response.
 - Plot the residuals against the fitted values. How are the zero response values manifested on the plot? What impact do these cases have on the interpretation of the plot?
 - Create a binary response variable which distinguishes zero values of `involact`. Fit a logistic regression model with this response but with the same five predictors. What problem occurred during this fit? Explain why this happened.
 - Fit a smaller model using only `race` and `age`. Interpret the z-statistics. Test for the significance of the two predictors using the difference-in-deviances test. Which test for the significance of the predictors should be preferred?
 - Make plot of `race` against `age` which also distinguishes the two levels of the response variable. Interpret the plot and connect it to the previous model output.
 - Refit the logit model but use a probit link. Compare the model output between the logit and probit models. Which parts are similar and which parts differ substantively? Plot the predicted values on the probability scale against each other and comment on what you see.

- (h) Which binary response model is most comparable to the Gaussian linear model? Contrast the drawbacks between the Gaussian and binary response models.
2. Aflatoxin B1 was fed to lab animals at various doses and the number responding with liver cancer recorded. The data may be found in the dataset `aflatoxin`.
- Plot the proportion of animals with liver cancer against the dose.
 - Fit a logistic regression for the number of animals with liver cancer as a function of the dose. Assess the statistical significance of the dose with two different tests. Which test is preferable?
 - For the fitted model, compute the predicted probability of liver cancer over the range of the data. Show the predicted curve on top of the data shown in (a).
 - Change to a probit link. Compute the predicted probability under this model and add the curve to the existing plot.
 - Compute the predicted probability of liver cancer for a dose of 25 ppb on the link scale. Compute a 95% confidence interval. Transform onto the probability scale.
 - Compute the predicted probability directly on the probability scale and report the standard error. Use this standard error to construct a 95% confidence interval for the probability. Which of the two methods of computing the CI as seen in this and the previous question is preferable?
 - Compute the effective dose at 1%, 10% and at 50% levels with associated standard error. Remark on any incongruity in your calculated effective doses.
3. The `infert` dataset presents data from a study of secondary infertility (failure to conceive after at least one previous conception). The factors of interest are induced abortions and spontaneous abortions (e.g., miscarriages). The study matched each case of infertility with two controls who were not infertile, matching on age, education and parity (number of prior pregnancies).
- Construct cross-classified tables by number of spontaneous and induced abortions separately for cases and controls. Comment on the differences between the two tables.
 - Fit a binary response model with only spontaneous and induced as predictors. Determine the statistical significance of these predictors. Express the effects of the predictors in terms of odds.
 - Fit a binary response model with only education, age and parity as predictors. Explain how the significance (or lack thereof) of these predictors should be interpreted.
 - Now put all five predictors in a binary response model. Interpret the results in terms of odds.
 - Fit a matched case control model appropriate to the data. Interpret the output and compare the odds to those found in the previous model.
 - The spontaneous and induced predictors could be viewed as ordinal due to the grouping in the highest level. Refit the model using ordinal factors rather than numerical variables for these two predictors. Is there evidence that the ordinal representation is necessary?

4. The data in `downs.bc` found in the `boot` package describes the proportion of births with Down's syndrome by the ages of the mothers collected in British Columbia.
- (a) Plot the proportion of Down's births against age and comment on the relationship.
 - (b) Fit a binomial response model for the Down's births with age as a linear predictor. Comment on the significance of the age effect.
 - (c) Display the predicted probability on the plot from (a). Comment on the quality of the fit.
 - (d) Add a quadratic term in age to the model and display the predicted probability on the same plot. Is the fit improved?
 - (e) Compute the predicted probability of a Down's birth at age 30 and at age 40 using the previous model. What is the odds ratio of these two probabilities? Now compute the predicted response on the link scale at age 30 and age 40. Exponentiate the difference between these two predictions. Explain why the answer is identical to that computed in the first part of this question.
 - (f) Fit a binomial response model with just a linear predictor in age but now using the complementary log-log link. Display the fitted probability curve on top of the observed proportions. Comment on the fit.
 - (g) Exponentiate the coefficient of age in the previous model. Now compute the predicted probability of a Down's birth at age 30 and at age 31. Compute the ratio of these two probabilities. Explain why this ratio is the same as the exponentiated coefficient. Give an interpretation in the context of this data.

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